

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Eighth Semester of B. Pharm. Examination
University Theory Examination May 2017
PH411 Pharmaceutical Analysis-IV

Date: 15.05.2017, Monday

Time: 01.30 p.m to 04.30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

- A (a) Write about magnetic properties of nuclei. 05
- (b) Enumerate factors affecting the chemical shift of the compound in NMR. Explain any two in detail. 05
- B (a) Give difference between ^1H -NMR and ^{13}C -NMR. 05
- (b) What is shielding and deshielding in NMR? Explain giving suitable example. 05
- C (a) Write a brief note on ICP-AES technique. 05
- (b) Draw a neat diagram of Hollow Cathod Lamp. Explain its working principle. 05

Q 2 Attempt any TWO of the following

- A (a) Why tetramethyl silane is used as a standard compound in ^1H NMR spectroscopy? Enlist solvents used in ^1H NMR. Define Coupling constant. 05
- (b) Solve the followings: 05
- (1) A molecule has molecular formula C_5H_{12} and ^1H NMR spectrum shows singlet at $\delta - 1.1$. Find the structure.
- (2) Predict the ^1H NMR chemical shift values and splitting patterns for 1,1-dichloroethane ; acetone
- B (a) Explain the principle of Atomic Absorptin Spectroscopy. 05
- (b) What types of interferences are generally encountered in Atomic Spectroscopy. Describe their remedies. 05

- C (a) Describe different types of atomization techniques used in Atomic Spectroscopy. Write their advantages and disadvantages. 05
- (b) Narrate the applications of Atomic Spectroscopic Techniques in Pharmaceutical Analysis. 05

SECTION- II

- Q 3** Attempt any TWO of the following
- A Discuss the theory and principle of Mass Spectrometry. 10
- B (a) Write a note on Mc Lafferty Rearrangement for MS. 05
- (b) Explain Bragg's law with a suitable diagram. 05
- C Differentiate 10
- (a) Hard and soft ionisation sources
- (b) Gas phase and desorption phase ionisation sources
- Q 4** Attempt any TWO of the following
- A Explain the principle of MALDI TOF with suitable diagram 10
- B Discuss the applications of X-ray diffraction method in Pharmacy. 10
- C Write in detail about Supercritical Fluid Chromatography. 10
-